Medical School Environments and Physician Treatment Styles

Shirley Cai[*] Ian McCarthy[†]

May 11, 2026

Abstract

We examine the role of medical school environments in shaping physicians' later treatment decisions, focusing on a specifc subset of cardiology in which physician discretion is particularly saliant and separate from patient need. Using a year-matched measure of cath lab availability in the hospital referral region of each cardiologist's medical school, we estimate that a 10-percentage-point increase in training-period availability raises later catheterization rates by 0.58 percentage points, identified across graduation cohorts within medical school HRR. The imprint is concentrated among interventional cardiologists and operates through measurable features of the training environment rather than ranking or prestige. Combining this training imprint with effects from their peers or general practice environments, identified from mid-career movers, yields a long-run multiplier of roughly 1.5 on per-cardiologist cath rates. The findings suggest that training-side interventions, if pursued at scale, would propagate through cohort transition into a meaningful reduction in place-based variation in physician treatment decisions.

[*]Emory University. shirley.cai@emory.edu

[†]Emory University & NBER. ian.mccarthy@emory.edu

1 Introduction

Physicians exercise meaningful discretion in their treatment decisions, and individual physicians vary substantially in how they exercise that discretion across clinically similar patients. This variation in physician treatment decisions is a well-documented contributor to geographic dispersion in health care utilization and spending, with as much as 60% of regional variation in spending attributable to such 'place-based' variation rather than to differences in patient need (Finkelstein et al., 2016; Molitor, 2018; Cutler et al., 2019; Badinski et al., 2023). These patterns are not arbitrary. They are shaped over the course of a long training pipeline, from medical school through residency and fellowship and into practice, and they continue to evolve through peer interactions and accumulated experience once physicians enter practice. Understanding which features of these environments shape later treatment style, and how those features propagate over a physician's career, is essential for designing supply-side interventions that aim to reduce inefficient variation in care.

In this paper, we exploit rich claims data in a care setting in which physician discretion is particularly salient, and we combine this with historic variation in clinically relevant technologies when physicians attended medical school. This allows us to credibly identify the effects of physicians' training environments on future practice patterns, separate from other concurrent place-based factors such as peers, arrangement with hospitals, and many others. Existing work has examined the residency and post-training environments in some detail, while the medical school environment has received less direct empirical attention. Part of the reason is measurement. The most common proxy in this literature, medical school rank (Schnell and Currie, 2018; Doyle Jr et al., 2010), bundles research intensity, faculty composition, student selectivity, and prestige, and it does not directly capture features of the medical school environment that would directly shape practice style. A more direct measure of the training environment, one that varies both across institutions and across graduation cohorts within institution, allows us to identify the medical school imprint on a specific clinical margin and quantify how it propagates through a physician's career.

Our analysis focuses on cardiologists treating non-ST-elevation myocardial infarction (NSTEMI) patients in Medicare claims data. NSTEMI is a useful empirical setting because the catheterization decision is meaningfully discretionary. Treatment guidelines support catheterization for many but not all NSTEMI patients, leaving substantial gray-zone cases in which physician practice style drives the decision (Chan et al., 2011; Amsterdam et al., 2014). We measure the training environment a medical student was exposed to as the share of hospitals in the medical school's hospital referral region (HRR) with a cardiac catheterization laboratory ("cath lab") in the year the student matriculated, drawn from the American Hospital Association's Annual Survey. This measure varies both across HRRs (some markets had cath labs earlier than others) and within HRR across graduation cohorts (cath labs proliferated steadily from 1980 to 2003), and the variation is plausibly independent of any individual medical student's future behavior or choice of medical school itself.[1]

We combine this training-side measure with a Medicare claims panel of cardiologists treating NSTEMI patients between 2008 and 2018. Our identification effectively rests on a mover design, albeit a mover from training to final practice rather than mid-career moves. Specifically, around 87% of cardiologists in our sample currently practice in a different HRR from the one in which they attended medical school, which generates the variation we exploit. Conditional on current practice HRR and year, cardiologists who trained in different HRRs were exposed to different training-period cath lab availability, and we estimate the imprint coefficient under a conditional-independence assumption that we defend with descriptive evidence on mover characteristics and origin-destination flows.[2] We further tighten the identification by adding fixed effects for the medical school HRR. The medical school imprint is then identified from cardiologists who attended medical school in the same HRR but in different graduation cohorts and thus encountered different cath lab availability during their clinical rotations.

We find that training-environment exposure has a causal effect on later catheterization rates. The cross-sectional mover specification produces an imprint coefficient of 0.035 (SE 0.011), which strengthens to 0.058 (SE 0.021) once we add medical school HRR fixed effects and identify the coefficient via cross-cohort variation within origin HRRs. The within-origin specification rules out fixed characteristics of medical school HRRs that might correlate with the type of physicians they produce. The effect is concentrated in interventional cardiologists, where the imprint coefficient is 0.058 (SE 0.019), compared to 0.013 (SE 0.019) for general cardiologists. The pattern is consistent with the procedural orientation a medical student observes during clinical rotations being most consequential for the subspecialty most directly engaged in catheterization. The estimates imply that a 10 percentage point increase in the training-period cath lab share, roughly the gap between the 25th and 75th percentile HRR in the mid-1990s, translates into a 0.58 percentage point increase in risk-adjusted catheterization rates (or about 0.04 standard deviations) when those same physicians are ultimately practicing cardiologists.

Subsequent analysis suggests that the medical school imprint operates through training-

[1] We consider questions of selection in more detail throughout our analysis as well as the supplemental appendix.

[2] The original move from medical school to first practice location precedes our 2008–2018 panel, so the identifying variation is cross-sectional rather than within-physician. We discuss the implications for our identification strategy in Section 3.

environment features rather simply reflecting medical school prestige. To examine this, we replace cath lab availability with a year-matched measure of NIH research-funding rank, computed from the NIH ExPORTER bulk files. This analysis shows effectively a null result for ranking during medical school on future catheterization rates. The cath lab availability measure retains its predictive power once rank is controlled, and the rank coefficient remains small and statistically insignificant across alternative parameterizations of rank (continuous, tiered, and binary top-25). These results affirm that our analysis identifies effects from procedurally rich training environments, separate from a medical school's overall ranking or prestige.

We further estimate how cardiologists adapt to their current practice environment after training. A within-physician event study around mid-career moves yields a sharp jump in catheterizations at the year of move, with no evidence of pre-trends, and a pooled post-move adaptation rate of roughly 35% of the destination gap. The destination response is asymmetric, with up-movers (cardiologists moving to higher catheterization markets) adapting more readily than down-movers (cardiologists moving to lower catheterization markets). This asymmetry is broadly consistent with cardiologists adjusting upward toward more intensive peers more readily than they adjust downward toward less intensive ones. Combining the destination response with our training imprint in a cohort-transition calibration produces a long-run multiplier of roughly 1.5 on per-physician practice intensity. This magnifier arrises from spillovers of the training environment effect onto future peers, as cohorts replace one another and the peer environment evolves. Incorporating such spillovers, our estimates suggest that a 10 percentage point increase in cath lab shares during training translates into a 0.05 standard deviation in risk-adjusted catheterization rates. This magnitude aligns with existing literature on other more targeted interventions, including audit-and-feedback and peer-comparisons (Ivers et al., 2012; Doctor et al., 2018; Barnett et al., 2023), but operates through a slower channel that compounds across cohorts.

Our analysis contributes to three distinct areas of economics and health policy. First, we contribute to the literature on sources of supply-side variation in health care utilization. Finkelstein et al. (2016) estimate that 50–60% of geographic variation in Medicare spending is attributable to supply-side factors rather than patient demand, with subsequent work attributing much of that share to physician practice style (Molitor, 2018; Cutler et al., 2019; Badinski et al., 2023). This literature has documented the magnitude of practice-style variation and traced its post-training evolution, but the upstream determinants of practice style at the start of a physician's career have received less direct attention. Our analysis identifies one such determinant, the procedural environment at the medical school during clinical rotations, and shows that the imprint persists into practice and is concentrated in the subspecialty most exposed to the procedural margin in question.

Second, we contribute to the literature on training as a determinant of physician behavior. Existing work has primarily relied on rank as a proxy for training quality (Schnell and Currie, 2018; Doyle Jr et al., 2010), with a related literature examining specific features of residency training (Epstein and Nicholson, 2009). Our analysis remains distinct from this literature in three key ways: 1) we replace rank with a direct measure of the procedural environment a medical student was exposed to, allowing us to identify a specific channel rather than a composite; 2) we exploit within-HRR variation in the procedural environment across graduation cohorts, which lets us rule out fixed characteristics of medical school markets that might correlate with the type of physicians they produce; and 3) we directly compare the procedural-exposure measure against a year-matched NIH-rank measure on the same sample and outcome, which allows us to adjudicate between the two channels rather than test them in isolation.

Third, we contribute to the literature on physician adaptation to current practice environments. Molitor (2018) shows that cardiologists who relocate adopt a substantial share of the gap between their old and new market's catheterization intensity. We complement this evidence using a within-physician event study that traces the adaptation trajectory in event time around mid-career moves and documents significant heterogeneity in the effects of the destination practice environment. Prior work further documents that physician treatment patterns respond to changes in reimbursement (Clemens and Gottlieb, 2014) and to evolving clinical evidence (DeCicca et al., 2024). We build on this work by separately identifying a training imprint and a destination response, then combining the two in a calibration that translates training-side effects into a long-run change in place-based intensity and variation.

Our findings are particularly salient given renewed policy interest in shaping the geographic distribution of physician practice through the medical training pipeline, including state-level loan-repayment and residency-expansion programs targeting underserved areas. If the medical school environment imprints durable practice-style features that persist into practice and propagate through peer environments via cohort turnover, then training-side interventions provide a slow but compounding lever on practice-style variation. The lever is partial. Training imprints are one input into practice style, and the within-physician adaptation we document indicates that cardiologists do continue to update in response to current peer behavior. Even so, the long-run multiplier we estimate suggests that training-side equalization, if pursued at scale, would propagate through cohort transition into a meaningful reduction in place-based variation in cardiac catheterization rates over the medium- to long-term.

2 Setting and Data

2.1 NSTEMI Catheterization and Cardiologist Training

Non-ST-elevation myocardial infarction is a heart attack characterized by partial occlusion of a coronary artery without persistent ST-segment elevation on the electrocardiogram.[3] NSTEMI patients are typically less acutely unstable than STEMI patients (for whom rapid catheterization is standard) but face elevated short-term mortality risk. Treatment guidelines support catheterization for many but not all NSTEMI patients, with the decision turning on a clinical judgment about the patient's risk profile, comorbidities, and likely benefit from invasive intervention. Substantial gray-zone cases exist in which the catheterization decision reflects clinical discretion rather than a firm clinical guidelines, which is what makes NSTEMI a useful setting for studying physician practice style. Variation in catheterization across cardiologists treating similar NSTEMI patients therefore plausibly reflects a physician's discretionary practice intensity rather than differential clinical indication or patient mix.

Cardiologists in the United States enter practice through a long, sequential training pipeline. Allopathic medical schools grant the MD degree over four years, with the final two years spent in clinical rotations at affiliated teaching hospitals.[4] Cardiology training continues with three years of internal medicine residency followed by three or more years of cardiology fellowship, with longer fellowships for interventional cardiology, electrophysiology, and advanced heart failure subspecialties. The procedural environment a medical student observes during clinical rotations is the foundation we have in mind when we measure training-period cath lab availability. For example, take a cardiologist for which we observe treating NSTEMI patients in 2010. Working backward from four years of medical school, three years of internal medicine residency, and three years of cardiology fellowship, that cardiologist matriculated medical school around 2000. For interventional cardiologists and advanced heart failure subspecialists, the additional fellowship year shifts the matriculation year to 1999, and for electrophysiologists, the two-year fellowship shifts it to 1998. Ultimately, the cath lab share we attribute to their training period is the share that prevailed in their medical school HRR in the implied year of matriculation.

[3]The ST segment refers to the portion of the electrocardiogram between the S and T waves. Persistent elevation in this segment typically indicates complete coronary occlusion and identifies an ST-elevation myocardial infarction (STEMI), which warrants immediate revascularization. The absence of persistent ST-segment elevation distinguishes NSTEMI cases, which present a wider range of underlying severity and a correspondingly wider range of appropriate treatment strategies (Amsterdam et al., 2014).

[4]In the U.S., there are both allopathic medical schools, which grant MDs, and osteopathic medical schools, which grant DOs. We focus on MDs because only roughly 11% of U.S. medical school graduates are DOs, with an even smaller share among cardiologists (AAMC, U.S. Physician Workforce Data Dashboard).

2.2 Data Sources

Medicare claims. Our analytical sample derives from 100% Medicare fee-for-service (FFS) data for inpatient and carrier claims from 2008 through 2018, accessed via the Centers for Medicare and Medicaid Services (CMS) Virtual Research Data Center (VRDC). NSTEMI admissions are identified by ICD-9 (410.71) and ICD-10 (I21.4) diagnosis codes on inpatient claims, retaining new episodes only. Cardiac catheterization, percutaneous coronary intervention, and coronary artery bypass graft procedures are identified from procedure codes within 90 days of the index NSTEMI admission. Carrier claims link each NSTEMI episode to the treating cardiologist, which we use to assign a single focal cardiologist per episode following the procedure described in Section 2.3. Beneficiary summary files provide demographics, fee-for-service coverage status, and the inputs to construct the 31 Elixhauser comorbidity flags from ICD diagnosis codes in the year prior to the NSTEMI episode.

MD-PPAS and Physician Compare. We identify cardiologists in the Medicare Data on Provider Practice and Specialty (MD-PPAS) file using primary-specialty fields for cardiology, interventional cardiology, clinical cardiac electrophysiology, and advanced heart failure and transplant cardiology. MD-PPAS also reports each cardiologist's primary practice ZIP code based on Medicare claim volume. We obtain medical school name, graduation year, and gender from the CMS Physician Compare National Downloadable File. We map medical schools to ZIP codes using a hand-curated crosswalk restricted to LCME-accredited U.S. allopathic schools, then assign each ZIP to a hospital referral region (HRR) using the Dartmouth Atlas ZIP-to-HRR crosswalk. We exclude cardiologists for whom the medical school field is missing or coded as "OTHER" (which captures most foreign-trained physicians and a small number of unmatched U.S. schools).

AHA Annual Survey. The AHA Annual Survey provides a rich set of hospital characteristics for essentially all non-federal general acute care hospitals from 1980 onward. We use the AHA to construct cath lab availability at the medical school HRR level.[5] We measure cath lab availability using a simple binary indicator of direct hospital provision of cardiac catheterization, and we aggregate to the HRR-year level by computing the share of hospitals for which the indicator is one. The aggregated AHA series exhibits a smooth secular trend across the 1994 questionnaire revision, indicating that the recoding does not introduce a measurement break. We further construct supplementary HRR-year measures of open-heart surgery (OHSRGHOS), cardiac intensive care (CICHOS), and adult cardiac services

[5] Cath lab availability derives from the AHA variable 'CCLABHOS' from 1994 onward, with the predecessor 'CCLAB82' covering 1980–1993.

(ACARDHOS) for use in robustness specifications.

NIH ExPORTER. To distinguish procedural exposure from medical school prestige, we construct a year-matched panel of medical school NIH research funding from the NIH ExPORTER bulk files. We aggregate annual project-level awards (1985–2025) to the medical school level and rank schools by total NIH funding within each year. The constructed series is validated against published Blue Ridge Institute for Medical Research (BRIMR) rankings, with correlation exceeding 0.99 in the overlap years.

2.3 Sample Construction and Key Variables

Cardiologist assignment and volume restriction. For each NSTEMI episode, we assign a single attending cardiologist following the procedure of Molitor (2018), prioritizing the first cardiologist who delivers a consultation, then the first cardiologist of record for inpatient care, and finally the first cardiologist seen by the patient on the carrier line. We exclude episodes for which no cardiologist could be assigned and patient-years for which Medicare FFS coverage was not present for the entire year. We restrict to cardiologist-years with at least 11 NSTEMI patients, which provides stable per-physician estimates of catheterization rates and aligns with the CMS cell-size requirements for data export. The trade-off of the volume restriction is that established, higher-volume cardiologists are over-represented in our sample relative to the broader cardiologist population. We discuss the implications for our event-study identification in Section 4.

Residualized catheterization rate. Our patient-level outcome is an indicator equal to one if the patient received a catheterization within two days of admission. We then residualize this indicator against the 31 Elixhauser comorbidities and year fixed effects in a linear probability model, retain the patient-level residual, and aggregate to the cardiologist-year as a volume-weighted mean. The residualization removes mechanical variation across cardiologists that would arise from differences in patient comorbidity mix, isolating the practice-style component.

Training-period cath lab share. For each cardiologist, we match the AHA HRR-year cath lab share at the start of medical school, defined as graduation year minus three. AHA cath lab data are available from 1980 through 2003, so the year-matching produces clean assignments for cardiologists with graduation years from 1983 through 2006. We restrict our analytical sample to this graduation-year window. Cardiologists who attended medical school in the same HRR but in different cohorts therefore receive different training-period values, which is the variation we exploit in our preferred within-origin specification.

Mover status and current practice environment. We assign each cardiologist's current-practice HRR using the Dartmouth ZIP-to-HRR crosswalk applied to the MD-PPAS practice ZIP. Movers are defined as cardiologists whose current practice HRR differs from their medical school HRR. To measure the current peer environment, we construct a leave-one-out HRR-year cath rate, defined as the volume-weighted mean residualized cath rate of all other cardiologists practicing in the cardiologist's HRR-year cell, excluding the focal cardiologist.

Practice characteristics. Cardiologists in our sample work in varied settings, ranging from large hospital-affiliated cardiology groups to smaller office-based practices, and these settings may themselves shape catheterization behavior independently of the training environment we are estimating. We construct two practice-setting characteristics from MD-PPAS to capture this dimension of the cardiologist's current work environment. The first is the share of the cardiologist's Medicare claims rendered at inpatient or hospital-outpatient places of service, which proxies for whether the practice is predominantly hospital-based versus office-based or ambulatory. The second is the log of the cardiologist's primary TIN-level Medicare patient volume, which proxies for the size of the practice group the cardiologist works within. Both vary at the cardiologist-year level. We use them as covariates in our preferred specifications to test whether the training imprint is sensitive to differences in current practice setting beyond what practice HRR fixed effects absorb.

2.4 Descriptive Statistics

The final analytical panel contains roughly 10,800 cardiologist-year observations covering about 3,200 unique cardiologists with non-missing training-period exposure, restricted to the 1983–2006 graduation-year window. Table 1 reports summary statistics at both the cardiologist-year and cardiologist levels. The mean residualized cath rate is approximately zero by construction, with a standard deviation of roughly 0.16. Average NSTEMI volume is 16 patients per cardiologist-year. Roughly three-quarters of the cardiologists in our sample are general cardiologists, with another 20% in interventional cardiology and the remainder split among electrophysiology and advanced heart failure.

Most cardiologists in our sample are cross-sectional movers in the sense that their current practice HRR differs from their medical school HRR. The mover share is roughly 87% by cardiologist-year. Table 2 compares movers and stayers on a set of pre-treatment characteristics. Movers come from medical school HRRs with somewhat lower mean cath lab share than stayers' medical school HRRs, suggesting that selection on training-environment intensity is at work. Movers also differ slightly on graduation cohort and subspecialty composition. We address selection via inverse probability weighting in Section 3.4.

Two features of the data help to inform our identification strategy. First, training-period cath lab availability varies substantially across HRRs, as do realized cath rates. Figure 1 maps the average residualized cath rate across HRRs in our sample. Clusters of high-cath and low-cath markets persist across years, consistent with the place-based variation in practice style that motivates the paper more generally. Second, movers face meaningful variation in the gap between their training environment and their current peer environment, which is the source of identifying variation in our destination event study. Figure 2 plots the distribution of this gap, which is roughly symmetric around zero with substantial mass in both tails. The symmetry indicates that movers in our sample do not concentrate in one direction of move, so we have variation to identify both up- and down-movers in the event study (Section 4). We discuss details of these analyses and identification strategies in their respective sections below.

3 Medical School Environments and Physician Treatment Styles

The descriptive evidence in Section 2.4 hints at a positive association between training-period cath lab availability and later catheterization rates. The raw correlation, however, cannot speak to whether the medical school environment causally shapes later practice style. Cardiologists who train in cath-rich HRRs may subsequently settle in cath-rich practice destinations, exhibit stronger procedural orientation that influenced their school choice in the first place, or attend higher-ranked schools whose prestige drives the pattern. This section uses a mover-design framework to isolate the medical school imprint on cardiologist catheterization decisions. We first lay out our identification strategy and present the headline estimates, then assess whether the imprint operates through procedural exposure rather than medical school ranking, examine subspecialty heterogeneity, and close with a battery of robustness checks.

3.1 Identification Strategy

Our initial identification strategy is a cross-sectional mover design that compares cardiologists who now practice in the same HRR but were trained in different medical school HRRs. If medical school training environments shape later practice intensity, we should observe that cardiologists trained in cath-rich HRRs catheterize at higher rates than cardiologists trained in cath-poor HRRs, even after we condition on where the cardiologist now practices. The condition on current practice HRR is essential. Without it, we would conflate the training imprint with the destination effect documented in Molitor (2018), since cardiologists who trained in cath-rich markets are also more likely to settle in cath-rich destinations. The estimating equation is

$$y_{it} = \beta_{\text{train}} \tau_i + \delta_{d(i,t)} + \gamma_t + \varepsilon_{it}, \tag{1}$$

where y_{it} is the volume-weighted mean residualized cath rate of cardiologist i in year t, τ_i is the training-period cath lab share in cardiologist i's medical school HRR (year-matched), $\delta_{d(i,t)}$ is a current-practice HRR fixed effect, and γ_t is a year fixed effect. We weight by NSTEMI volume and cluster standard errors at the medical school HRR level.

Under the assumption that cardiologists do not select their practicing HRR based on procedural exposure during their medical school years, this design aims to remove correlation between training-period cath lab share and unobserved determinants of current practice intensity. A violation would require cardiologists to sort into destinations whose unobserved characteristics correlate with their own training-period cath lab exposure. If such selection were present, it would likely bias our estimates upward, as cardiologists with higher procedural orientation would settle in markets with characteristics that further reinforce that orientation. To assess the reasonableness of this assumption, Figure 3 shows that cardiologists move to a wide spread of destinations rather than concentrating on one or two. Movers from the median origin HRR are observed in 23 distinct destination HRRs, with a destination-share HHI of 0.066, supporting the view that destinations are not driven by training-side composition.

One issue with this strategy is that it leaves untouched any time-invariant feature of the medical school HRR that might generate the imprint we estimate. Two medical schools could differ in cath lab availability and also differ in the prestige of their teaching hospitals, the patient case mix during clerkship rotations, the typical career path of their graduates, the local professional culture, or the financial incentives faced by faculty mentors. Any of these other features could be what shapes later practice intensity, with cath lab availability simply correlated with the true mechanism. We therefore also consider a cross-cohort identification strategy that adds fixed effects for the medical school HRR:

$$y_{it} = \beta_{\text{train}}\tau_i + \rho_{m(i)} + \delta_{d(i,t)} + \gamma_t + \varepsilon_{it}, \tag{2}$$

where $\rho_{m(i)}$ is a medical school HRR fixed effect. Under this strategy, β_{train} is identified from cardiologists who attended medical school in the same HRR but in different graduation cohorts and thus encountered different cath lab availability during their clinical rotations. The national mean cath lab share rose from roughly 0.14 in 1980 to roughly 0.39 in 2003, with substantial variation across HRRs in the timing of the rollout. The identifying assumption embedded in equation (2) is that within-HRR cath lab adoption is uncorrelated with cohort-level shifts in other determinants of later practice intensity. We caution that cohorts entering medical school in the early 1980s and the early 2000s differ on margins beyond local cath lab proliferation, including managed care diffusion, the introduction of stenting and drug-eluting stents, and curriculum shifts. We revisit this concern in the robustness analysis below.

Results from both strategies are reported in Table 3. Panel A reports the cross-sectional design from equation (1), with destination HRR and year fixed effects. Panel B adds medical school HRR fixed effects per equation (2). Within each panel, column (1) is the baseline, column (2) adds physician characteristics (gender and years since graduation), and column (3) further adds practice characteristics (hospital-based share of Medicare claims and log of TIN-level patient volume). We deliberately exclude subspecialty from the set of controls, as subspecialty selection is itself a downstream outcome of the training environment we are estimating. We instead consider subspecialty as a dimension of heterogeneity in Section 3.3.

From column (1) of Panel A, the training-imprint coefficient is 0.035 (SE 0.011) and statistically distinguishable from zero at the 1% level. A cardiologist who trained in an HRR where every hospital had a cath lab catheterizes about 3.5 percentage points more often than a cardiologist trained in an HRR with no cath labs, holding current practice HRR and year fixed. Incorporating medical school HRR fixed effects in column (1) of Panel B increases our estimate of $\hat{\beta}_{\text{train}}$ to 0.058 (SE 0.021), suggesting that fixed origin-HRR features were attenuating the cross-sectional estimate. The training imprint is essentially stable across columns within each panel. Adding physician characteristics in column (2) and practice characteristics in column (3) leaves the coefficient in the same range, with the Panel B estimate increasing from 0.058 to 0.071 across the three columns. The pattern indicates that the imprint is not driven purely by sorting on observable cardiologist or practice characteristics. Ultimately, we treat the within-origin baseline in column (1) of Panel B as our preferred specification. For this estimate, the identifying variation is the difference in cath lab exposure between cardiologists from the same medical school but different graduation cohorts, which is plausibly orthogonal to cardiologist-level unobservables in a way that cross-sectional variation alone is not.

3.2 Training Environment vs. Medical School Ranking

While Table 3 suggests that medical school training environment has an economically meaningful effect on future treatment patterns, an alternative interpretation is that what really matters is the prestige of the medical school, with cath lab availability simply serving as a proxy for overall school quality. Indeed, several papers in the practice-style literature use medical school rank as a proxy for training quality (Schnell and Currie, 2018; Doyle Jr et al., 2010). While rank conceptually captures potentially different dimensions of medical school characteristics (e.g., research intensity, faculty composition, student selectivity, and prestige), the two channels could in principle both matter, or the effects of one could be absorbed by the other. We test this by replacing τ_i in equation (2) with the cardiologist's NIH-funding rank tier from our year-matched ExPORTER panel.

Table 4 reports the rank specifications. The continuous $-\log(\text{rank})$ coefficient is essentially zero and statistically indistinguishable from zero ($p = 0.28$). Tier indicators (Top 10, Top 11–25, Top 26–50, Top 51–100, with unranked as the reference) are likewise not statistically distinguishable from zero. The pattern is robust to restricting to movers and to alternative tier cutoffs. While rank may ultimately affect physician behaviors in other ways, it does not show up as a meaningful factor for physician catheterization rates of NSTEMI patients.

The evidence is therefore consistent with the medical school environment shaping later cath behavior through training-environment features rather than through the prestige or research orientation of the school itself. A cardiologist trained at a non-elite medical school in a cath-rich HRR carries the imprint as cleanly as one trained at an elite school in a cath-rich HRR, while training at a top-ranked school in a cath-poor HRR does not produce an imprint of comparable magnitude. We do not attempt to disentangle the specific channel through which a procedurally rich training environment imprints on later practice style, since our setting cannot separately identify direct-observation channels (e.g., medical students observing catheterization decisions during clinical rotations) from broader training-environment channels (e.g., regional norms among teaching faculty or peer-effect spillovers within the cardiology workforce a graduate later joins).

3.3 Subspecialty Heterogeneity

The training-environment interpretation makes a sharp prediction about subspecialty heterogeneity. If exposure to a procedurally rich training environment imprints a procedural orientation that persists into practice, then the imprint should manifest most strongly in cardiologists who later select into the most procedurally focused subspecialty. Interventional cardiologists perform catheterizations themselves, general cardiologists primarily make referral and management decisions, and electrophysiologists and advanced heart failure cardiologists perform other types of procedures and rarely catheterize NSTEMI patients directly. We test the prediction by stratifying the within-origin specification by cardiology subspecialty.

Table 6 reports the within-origin training-imprint estimate stratified by subspecialty. Among interventional cardiologists, $\hat{\beta}_{\text{train}} = 0.058$ (SE 0.019, $p = 0.002$). Among general cardiologists, the point estimate is 0.013 (SE 0.019) and statistically indistinguishable from zero.[6] Electrophysiologists and advanced-heart-failure cardiologists have small samples (358 and 65 cardiologist-years respectively) and imprecise estimates. The interventional concentration is consistent with the training-environment interpretation. Cardiologists who selected into the most procedurally oriented subspecialty bear the strongest imprint, while generalists, whose role is less procedurally focused, do not.

We caution that subspecialty selection is endogenous. Cardiologists with strong procedural orientation may sort into interventional cardiology, in which case the subspecialty heterogeneity reflects both a within-subspecialty practice-style effect and a between-subspecialty selection effect. Both interpretations are consistent with the broader claim that the imprint operates through procedural orientation rather than through generic medical school quality.

3.4 Robustness

We assess two threats to the within-origin training-imprint estimate.

The first is selection into being a mover, where we define a mover as a cardiologist whose current practice HRR differs from their medical school HRR (about 87% of our analytical sample). Table 2 shows that movers and stayers differ on observable characteristics, including graduation cohort, gender, subspecialty composition, and training-period cath lab share. The threat is that the same characteristics also correlate with unobserved practice-style propensities, in which case our cross-sectional mover variation reflects selection rather than a causal training effect. To address this, Table 7 reports an inverse-probability-weighted (IPW) version of the within-origin specification. We estimate a logit propensity score predicting

[6] The corresponding minimum detectable effect at 80% power is approximately 0.053, so the general-cardiology null is informative against effects of magnitude similar to the interventional coefficient.

mover status from the observable cardiologist characteristics listed above. We then weight each cardiologist by the inverse of their predicted probability of being in their observed mover-or-stayer group, which re-balances the sample so that movers and stayers have the same distribution of observable characteristics. The IPW-weighted training-imprint coefficient is 0.057 (SE 0.021), essentially identical to the unweighted baseline of 0.058. Observable selection on mover status does not drive the imprint.

The second is unobserved cohort-level variation within medical school HRR. Our within-origin specification absorbs fixed medical school HRR features but not features that shift over time within an HRR. Cohorts entering medical school in the early 1980s and the early 2000s differ along many margins beyond local cath lab proliferation, including managed care diffusion, the introduction of stenting and drug-eluting stents, and curriculum shifts. If any of these cohort-era shifts are correlated with cath lab proliferation within HRRs, our estimate could be picking up the era rather than the cath lab availability itself. To tighten the identification further, we add medical school HRR $\times$ graduation-decade fixed effects, which restrict the identifying variation to within-decade-within-HRR differences in training-period cath availability. The resulting coefficient is 0.067 with a wider standard error of 0.051, reflecting the more limited variation that remains. The imprint survives the absorption with similar magnitude, even as the precision drops.

4 Adaptation to Current Practice Environments

4.1 Motivation

Section 3 establishes that medical school environments leave a durable imprint on cardiologist cath behavior, but it does not directly address how the imprint interacts with the current peer environment. Molitor (2018) provides the closest existing evidence on this margin through cardiologist relocation, finding that cath rates adjust toward destination markets at a substantial rate. Identifying the destination response is necessary for two reasons. First, the cohort-transition calibration in Section 5 requires both the training imprint β_{train} and the destination response β_{dest} to characterize how training-side shifts propagate over time. Second, the within-physician estimate disciplines the cross-sectional triangulation in Section 4.5 and lets us quantify the role of unobserved selection in driving the cross-sectional gap. Our event study revisits this question within our NSTEMI sample and further examines whether the destination response is symmetric in direction, which speaks to whether training imprint and current peer environments act as substitutes or complements.

4.2 Mid-Career Mover Sample

A clean within-physician estimate of the destination response requires cardiologists who change practice HRRs during our panel. We define mid-career movers as cardiologists with at least two distinct practice HRRs in the 2008–2018 panel. Roughly 580 cardiologists in our analytical sample meet this criterion. Compared with the non-movers, mid-career movers are disproportionately interventional cardiologists (29% of movers vs. 22% of non-movers), have somewhat higher NSTEMI volume per year, are slightly younger by graduation cohort, and come from training HRRs with marginally higher cath lab share. Table 8 reports the comparison.

The mid-career mover sample is selected on observables relative to the broader analytical panel, and the within-physician event-study estimate of the destination response should be interpreted as a local effect within this subpopulation rather than a population parameter. The sample is also small, with roughly 580 cardiologists and a corresponding limit on the precision with which we can characterize event-time dynamics or stratify by subgroup. Across the destinations they go to, however, mid-career movers do not exhibit strong sorting on training intensity. Cross-sectional regression of move direction on training cath share is not statistically distinguishable from zero, suggesting that high-trained and low-trained movers go to similar mixes of destinations.

4.3 Event-Study Design

For each mid-career mover, define the move year as the first year in which their practice HRR differs from their initial-panel HRR. Define event time τ as calendar year minus move year. The event-study specification is

$$y_{it} = \alpha_i + \gamma_t + \sum_{\tau \neq -1} \beta_\tau \cdot \mathbf{1}[\tau_{it} = \tau] \cdot \Delta_i + \varepsilon_{it}, \tag{3}$$

where α_i is a cardiologist fixed effect, γ_t is a calendar-year fixed effect, Δ_i is the gap between the leave-one-out destination peer cath rate at the move year and the corresponding origin peer cath rate the year before the move, and β_τ traces the share of Δ_i reflected in cardiologist i's cath rate at event time τ, normalized to zero at $\tau = -1$. We restrict to the event-time window $[-3, +5]$ to retain a balanced panel of physician-years, weight by NSTEMI volume, and cluster standard errors at the cardiologist level.

4.4 Headline Adaptation

Figure 4 plots the event-study coefficients with 95% confidence intervals. Pre-move coefficients ($\tau = -3, -2$) are statistically indistinguishable from zero, consistent with the parallel-trends assumption. At the move year, $\hat{\beta}_0 = 0.443$ (95% CI [0.25, 0.63]), indicating immediate adaptation of roughly 44% of the peer-environment gap. Post-move coefficients persist, with the pooled post-move coefficient at 0.350.

The point estimates are smaller than the roughly 0.6 reported in Molitor (2018) for cardiologists generally, though the comparison is not apples-to-apples. Molitor's outcome is a broader practice-intensity index built from the cardiologist's procedure mix, while ours is the residualized NSTEMI cath rate, a narrower measure. Our sample is also restricted to high-volume cardiologists by the volume threshold described in Section 2. Within these restrictions, the event-study estimate is comparable in identification logic to Molitor's pooled adaptation rate.

4.5 Cross-Sectional Triangulation

We can also identify the destination response cross-sectionally, exploiting variation in destinations across cardiologists from the same medical school HRR. The specification is equation (2) with the destination peer cath rate (leave-one-out) added as a regressor,

$$y_{it} = \beta_{\text{train}} \cdot \tau_i + \beta_{\text{dest}} \cdot \pi_{d(i,t)} + \rho_{m(i)} + \gamma_t + \varepsilon_{it}.$$

The cross-sectional $\hat{\beta}_{\text{dest}}$ is 0.58, materially larger than the within-physician event-study estimate of 0.35. The gap is consistent with cross-sectional sorting of cardiologists into HRRs whose peer environments correlate with the cardiologists' own unobserved practice propensity. The within-physician identification removes that selection by holding the cardiologist fixed.

The cross-sectional estimate is also robust to inverse probability weighting on observable cardiologist characteristics, indicating that the gap between 0.58 and 0.35 is not driven by selection that standard IPW methods can correct. We treat the within-physician 0.35 as the cleaner causal estimate and use it for the policy calibration in Section 5, while reporting the cross-sectional 0.58 to triangulate against the prior literature.

4.6 Direction-Asymmetric Malleability

The pooled response averages over two directions of move that need not produce symmetric responses. Up-movers (cardiologists whose destination peer cath rate exceeds their origin peer rate) and down-movers (the reverse) follow different post-move trajectories. If training imprint and current environment act as complements, the two directions should produce symmetric responses. If they act as substitutes, cardiologists from cath-rich training environments should resist downward pressure from a less procedurally aggressive destination, and cardiologists from cath-poor training environments should adapt more fully to a more procedurally aggressive destination.

Figure 5 stratifies the event study by move direction. At the move year, up-movers and down-movers have similar response magnitudes (0.45 and 0.50 respectively, both statistically distinguishable from zero). Post-move trajectories diverge sharply. Up-movers' adaptation grows over time, with $\hat{\beta}_5 \approx 0.83$. Down-movers' adaptation fades, with $\hat{\beta}_5 \approx 0.16$.

The asymmetry is consistent with training imprint and current peer environment acting as substitutes rather than complements. Cardiologists exposed to a high-cath training environment retain that imprint and revert toward it after an initial accommodation to a lower-cath destination. Cardiologists from lower-cath training environments are more malleable to higher-cath destinations and continue to adapt over years. The asymmetry is identified within-physician and is not driven by sorting on training intensity, since high-trained and low-trained movers go to similar mixes of destinations (Section 4.2). The asymmetric pattern has implications for how training-side interventions propagate through the system over time, which we develop in Section 5.

5 Long-Run Policy Implications

5.1 Peer-Effect Amplification

Sections 3 and 4 jointly identify two parameters governing how training-side shifts propagate to current cath behavior. The training imprint $\beta_{\text{train}} = 0.058$ measures how a marginal increase in training-period cath lab share translates into an increase in own cath rate, identified within-origin across cohorts. The destination response $\beta_{\text{dest}} = 0.35$ measures how an increase in current peer-environment cath rate translates into an increase in own cath rate, identified within-physician. The two parameters compose. Suppose training-period cath lab share is uniformly raised by $\Delta\tau$ for all new cohorts of cardiologists entering an HRR. Each new cohort enters with practice intensity raised by $\beta_{\text{train}} \cdot \Delta\tau$ relative to baseline. As new cohorts enter and old cohorts retire, the HRR's peer-environment cath rate gradually shifts upward, and existing cardiologists adjust toward the new peer mean at rate β_{dest}. Their adjustment in turn contributes to further shifts in the peer mean for the next cohort. In steady state, the per-cardiologist long-run effect of the uniform training shift is

$$\Delta y^* = \frac{\beta_{\text{train}} \cdot \Delta\tau}{1 - \beta_{\text{dest}}}. \tag{4}$$

With $\beta_{\text{train}} = 0.058$ and the within-physician estimate $\beta_{\text{dest}} = 0.35$, the amplification multiplier $1/(1 - \beta_{\text{dest}})$ is approximately 1.54. A uniform 10-percentage-point shift in training-period cath lab share produces a long-run shift of roughly 0.89 percentage points in per-cardiologist cath rate, decomposing into 0.58 percentage points of direct training effect on the new cohorts and 0.31 percentage points of cross-cardiologist spillover that accumulates as new cohorts displace old ones and the peer environment evolves.

The multiplier is sensitive to which destination-response estimate we plug in. The cross-sectional analog $\beta_{\text{dest}} = 0.58$ produces a multiplier of roughly 2.38 and a long-run shift of about 1.38 percentage points for the same 10-percentage-point training shock. The within-physician estimate strips out the cross-sectional sorting discussed in Section 4.5 and is the cleaner causal parameter, but it is identified within a small selected sample of mid-career movers. The cross-sectional estimate covers a broader cardiologist population but inherits the sorting bias. We read the two estimates as bounding the true population parameter, with the multiplier falling somewhere between 1.5 and 2.4. The qualitative implications for the policy discussion below are unchanged across the bound.

Peer effects, identified in prior work as a feature of how individual cardiologists respond to their current environment, also serve as the mechanism through which training-side shifts propagate across cohorts over time. The training imprint and the peer-effect channel are complementary rather than competing, with peer-effect spillovers contributing roughly half to one and a half times as much to the long-run effect as the direct training imprint alone. We caution that the cross-cardiologist amplification embedded in the multiplier assumes incumbents adjust to peer-environment shifts at the same rate as the mid-career movers from whom we identify β_{dest}, which is most defensible at long horizons since incumbents eventually become the peer environment for newly arriving cohorts.

5.2 Variance Reduction Transition Path

The amplification in equation (4) characterizes the steady-state response to a uniform training shift, but it does not describe the speed of adjustment. To trace the speed, we simulate a cohort-by-cohort transition assuming 5% of cardiologists are replaced each year (corresponding to a 20-year career length). The simulation is illustrative rather than predictive. It tracks how an initial training-side equalization across HRRs would propagate through cohort turnover into cross-HRR variance in the cath rate under our reduced-form parameter estimates and a stated value of the share parameter s described below. A natural counterfactual asks what fraction of geographic variation in cath rates would dissipate if training-period cath lab availability were equalized across all medical school HRRs at the national mean. The answer depends on what share of cross-HRR cath-rate variance is attributable to the training-and-peer channel rather than to other place-based factors (hospital-system ownership, financial incentives, technology endowments, malpractice climate, local guideline adherence) that our equalization counterfactual leaves untouched.

We do not identify this share directly. Our mid-career mover sample is too small for an Abowd-Kramarz-Margolis-style two-way fixed-effect decomposition of geographic variance into physician and place components, and our reduced-form coefficients do not partition variance across the full set of channels. We instead anchor the share to the broader literature. Cutler et al. (2019) find that physician beliefs explain on the order of 60% of regional variation in end-of-life spending and a similar but somewhat smaller share for heart-attack spending. Badinski et al. (2023) report a place share of total geographic variation in health care utilization that, combined with Cutler et al.'s within-place physician share, suggests training-and-peer factors account for somewhere between 30% and 70% of place-based variation in cath-related outcomes. The remaining place-based variation reflects the other factors above.

The transition speed is governed by cohort replacement plus peer-effect propagation. With 5% annual turnover, half of the long-run variance reduction is realized within roughly a decade of the equalization shock, and the system approaches steady state by year 20. The total reduction in place-based variance under equalization is approximately s, where s is the training-and-peer share of place-based variance. Patient-attributable variation, which reflects real differences in clinical need, is unaffected by the counterfactual, which is appropriate because patient-need differences across geographic areas are not what supply-side reform should target.

5.3 Direction-Asymmetric Persistence

The pooled β_{dest} obscures the direction-asymmetric malleability documented in Section 4.6. Up-shocks to training cath lab availability propagate more durably than down-shocks. New cohorts entering at higher training intensity become incumbents whose elevated cath rates raise the peer mean, which then influences successive cohorts at the up-mover rate ($\hat{\beta}_5 \approx 0.83$ at long horizons). New cohorts entering at lower training intensity instead find themselves in higher-than-training peer environments and adapt upward toward the existing higher peer mean at roughly the same rate, partially undoing the policy.

The implication is that training-side reform is a more effective tool for raising practice intensity than for lowering it. A policy that increases procedural exposure at low-availability training HRRs would propagate forward, while a policy that decreases procedural exposure at high-availability training HRRs would dissipate as the next generation of cardiologists adapted upward to the existing peer environment. The asymmetry runs in a direction that is unhelpful for any clinical setting in which the policy goal is to reduce overuse, and it is not visible in the pooled estimates alone. We further caution that our calibration is per-cardiologist rather than per-HRR; translating to national policy magnitudes would require multiplying by physician counts and patient volumes, which we do not undertake here.

6 Conclusion

This paper examines whether the medical school environment in which a cardiologist trained leaves a causal imprint on their later catheterization decisions, and how that imprint interacts with the current peer environment to shape practice style over a career. Using a year-matched measure of cath lab availability in the cardiologist's medical school hospital referral region, drawn from the AHA Annual Survey, we identify the training imprint within-origin across graduation cohorts. We further estimate the destination response through a within-physician event study around mid-career moves and combine the two parameters in a cohort-transition calibration.

We find that training-environment exposure has a causal effect on later catheterization rates, with a within-origin coefficient of 0.058 (SE 0.021). The imprint operates through training-environment features rather than through medical school prestige; a year-matched NIH-funding rank measure shows no analogous gradient. The effect is concentrated in interventional cardiology, the subspecialty most directly engaged in the procedural margin we study. The destination response, identified within-physician, is roughly 0.35 in the pooled specification with a sharp jump at the move year and parallel pre-trends. The pooled response masks direction-asymmetric malleability, with up-movers adapting to roughly 83% of the destination gap by five years post-move while down-movers adapt to only 16%.

Our analysis aligns most directly with Molitor (2018), who identifies the destination-response parameter through cardiologist relocation. We complement that evidence in three ways: 1) we identify a separate training-imprint parameter using a year-matched AHA cath lab measure that is independent of physician outcomes; 2) we trace the destination response in event time around mid-career moves, which lets us document direction-asymmetric malleability that the pooled cross-sectional estimate obscures; and 3) we combine the destination response with the training imprint in a cohort-transition calibration, which provides a quantitative bridge between the two parameters that the existing literature has typically estimated in isolation.

We conclude with three policy observations. First, training-side reform is a slow but compounding lever on practice-style variation. The peer-effect spillover amplifies the direct training imprint by roughly half to one and a half times through cohort turnover, with the bulk of the long-run effect realized over horizons of two to three decades. Second, the asymmetry in destination response runs in a direction that is unhelpful for clinical settings in which the policy goal is to reduce overuse. A policy that increases procedural exposure at low-availability training HRRs would propagate forward, while a policy that decreases procedural exposure at high-availability training HRRs would dissipate as new cohorts adapted upward to the existing peer environment. Third, disciplining the share of place-based variation attributable to training-and-peer factors is a fruitful direction for future work. Our reduced-form coefficients identify the per-cardiologist response of cath rates to training-side shifts, but the share of geographic variation that training equalization can ultimately reduce depends on the relative contribution of training-and-peer factors versus other place-based channels (hospital-system ownership, financial incentives, technology endowments) that our equalization counterfactual leaves untouched. Settings with richer mover samples or richer institutional measurement of teaching-hospital affiliations could deliver tighter bounds on this share than our cardiologist mover sample supports.

References

Amsterdam, Ezra A., Nanette K. Wenger, Ralph G. Brindis, Donald E. Casey, Theodore G. Ganiats, David R. Holmes, Allan S. Jaffe, Hani Jneid, Rosemary F. Kelly, Michael C. Kontos, Glenn N. Levine, Philip R. Liebson, Debabrata Mukherjee, Eric D. Peterson, Marc S. Sabatine, Richard W. Smalling, and Susan J. Zieman, "2014 AHA/ACC Guideline for the Management of Patients With Non–ST-Elevation Acute Coronary Syndromes," *Circulation*, December 2014, *130* (25), e344–e426.

Badinski, Ivan, Amy Finkelstein, Matthew Gentzkow, and Peter Hull, "Geographic Variation in Healthcare Utilization: The Role of Physicians," October 2023.

Barnett, Michael L., Andrew Olenski, and Adam Sacarny, "Common Practice: Spillovers from Medicare on Private Health Care," *American Economic Journal: Economic Policy*, August 2023, *15* (3), 65–88.

Chan, Paul S., Manesh R. Patel, Lloyd W. Klein, Ronald J. Krone, Gregory J. Dehmer, Kevin Kennedy, Brahmajee K. Nallamothu, W. Douglas Weaver, Frederick A. Masoudi, John S. Rumsfeld, Ralph G. Brindis, and John A. Spertus, "Appropriateness of Percutaneous Coronary Intervention," *JAMA*, July 2011, *306* (1), 53–61.

Clemens, Jeffrey and Joshua D Gottlieb, "Do Physicians' Financial Incentives Affect Medical Treatment and Patient Health?," *American Economic Review*, 2014, *104* (4), 1320–1349.

Cutler, David, Jonathan S. Skinner, Ariel Dora Stern, and David Wennberg, "Physician Beliefs and Patient Preferences: A New Look at Regional Variation in Health Care Spending," *American Economic Journal: Economic Policy*, February 2019, *11* (1), 192–221.

DeCicca, Philip, Maripier Isabelle, and Natalie Malak, "How Do Physicians Respond to New Medical Research?," July 2024.

Doctor, Jason N., Andy Nguyen, Roneet Lev, Jonathan Lucas, Tara Knight, Henu Zhao, and Michael Menchine, "Opioid prescribing decreases after learning of a patient's fatal overdose," *Science*, August 2018, *361* (6402), 588–590.

Epstein, Andrew J. and Sean Nicholson, "The formation and evolution of physician treatment styles: An application to cesarean sections," *Journal of Health Economics*, December 2009, *28* (6), 1126–1140.

Finkelstein, Amy, Matthew Gentzkow, and Heidi Williams, "Sources of Geographic Variation in Health Care: Evidence From Patient Migration," *The Quarterly Journal of Economics*, 2016, *131* (4), 1681–1726.

Ivers, Noah, Gro Jamtvedt, Signe Flottorp, Jane M Young, Jan Odgaard-Jensen, Simon D French, Mary Ann O'Brien, Marit Johansen, Jeremy Grimshaw, and Andrew D Oxman, "Audit and feedback: effects on professional practice and healthcare outcomes," *The Cochrane Database of Systematic Reviews*, June 2012, *2012* (6), CD000259.

Jr, Joseph J Doyle, Steven M Ewer, and Todd H Wagner, "Returns to physician human capital: Evidence from patients randomized to physician teams," *Journal of health economics*, 2010, *29* (6), 866–882.

Molitor, David, "The evolution of physician practice styles: evidence from cardiologist migration," *American Economic Journal: Economic Policy*, 2018, *10* (1), 326–56.

Schnell, Molly and Janet Currie, "Addressing the Opioid Epidemic: Is There a Role for Physician Education?," *American Journal of Health Economics*, August 2018, *4* (3), 383–410.

Tables and Figures

Table 1. **Summary Statistics**[a]

	Mean	SD	Min	Max	N
Cardiologist-year level					
Residualized cath rate	0.013	0.161	-0.680	0.514	31,235
NSTEMI volume per year	15.9	5.8	11.0	133.0	31,235
Med school HRR intensity	0.025	0.036	-0.255	0.188	15,305
Destination HRR LOO intensity	0.015	0.079	-0.511	0.422	31,022
Δ intensity (dest $-$ med school)	-0.007	0.077	-0.448	0.367	15,210
Cardiologist level (per NPI)					
Graduation year	1989.8	10.7	1947.0	2012.0	9,495
Years observed in panel	3.3	2.5	1.0	10.0	9,568
Female	0.077				9,568
General Cardiology	0.756				9,568
Interventional Cardiology	0.201				9,568
Electrophysiology	0.036				9,568
Adv. Heart Failure	0.007				9,568

[a]Cardiologist-year observations from the 2008–2018 Medicare fee-for-service panel, restricted to cardiologists treating at least 11 NSTEMI patients in a given year and to graduation cohorts 1983–2006. The residualized cath rate is the volume-weighted mean of patient-level residuals from a linear probability model regressing an indicator for catheterization within two days of NSTEMI admission on the 31 Elixhauser comorbidities and year fixed effects.

Table 2. **Movers vs. Stayers**[a]

	Movers	Stayers	Diff.	p-value
Graduation year	1989.4	1990.0	-0.6	0.003
Female (%)	7.5	7.9	-0.3	0.530
General Cardiology (%)	74.1	76.8	-2.7	0.002
Interventional Cardiology (%)	21.1	19.3	1.8	0.035
Electrophysiology (%)	4.2	3.1	1.1	0.006
Adv. Heart Failure (%)	0.6	0.8	-0.1	0.474
Med school HRR intensity	0.022	0.036	-0.013	0.000
Years in panel	3.2	3.3	-0.1	0.087
Mean NSTEMI volume per year	14.2	14.6	-0.4	0.000
First observation year	2011.5	2011.8	-0.3	0.000
Cardiologists	4,121	5,447		

[a] Cardiologist-level comparison of movers (whose current practice HRR differs from their medical school HRR) and stayers. Means are reported by group, with the difference in means and a p-value from a two-sample t-test. The training cath share is the share of hospitals in the cardiologist's medical school HRR with a cardiac catheterization laboratory in the year of matriculation, drawn from the AHA Annual Survey.

Table 3. **Effect of Training Environment on Residualized Catheterization Rate**[a]

	Baseline	+ physician vars	+ practice vars
Panel A. Destination FE			
Training-HRR cath lab share	0.035***	0.025*	0.025*
	(0.011)	(0.013)	(0.014)
Current-HRR cath culture (LOO)	0.047	0.046	0.050
	(0.030)	(0.031)	(0.031)
Observations	10,729	10,729	10,729
Panel B. Origin + Destination FE			
Training-HRR cath lab share	0.058***	0.062*	0.071**
	(0.021)	(0.034)	(0.034)
Current-HRR cath culture (LOO)	0.049	0.048	0.054*
	(0.030)	(0.031)	(0.030)
Observations	10,729	10,729	10,729

[a]The outcome is the volume-weighted mean residualized cath rate at the cardiologist-year level. Panel A reports the cross-sectional mover specification with destination HRR and year fixed effects (equation 1). Panel B adds medical school HRR fixed effects (equation 2). Column (1) is the baseline. Column (2) adds physician characteristics (gender and years since graduation). Column (3) further adds practice characteristics (hospital-based share of Medicare claims and log of TIN-level Medicare patient volume). All specifications weight by NSTEMI volume and cluster standard errors at the medical school HRR level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 4. **Effect of School Rank on Residualized Catheterization Rate**[a]

	(1)	(2)	(3)
Panel A. $-\log$*(NIH rank)*			
$-\log$(NIH rank)	0.006**	0.005*	0.005
	(0.003)	(0.003)	(0.003)
Training cath lab share		0.058***	0.066*
		(0.022)	(0.035)
Panel B. NIH rank tier indicators (ref. = unranked)			
Top 51-100	0.001	0.003	0.000
	(0.012)	(0.012)	(0.011)
Top 26-50	-0.003	-0.002	-0.005
	(0.015)	(0.015)	(0.015)
Top 11-25	0.013	0.015	0.012
	(0.013)	(0.012)	(0.013)
Top 10	0.014	0.015	0.011
	(0.015)	(0.015)	(0.015)
Training cath lab share		0.059***	0.066*
		(0.021)	(0.035)
Physician characteristics	No	No	Yes
Practice characteristics	No	No	Yes
Med school HRR FE	Yes	Yes	Yes
Practice HRR FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	10,609	10,609	10,609

[a]The outcome is the volume-weighted mean residualized cath rate at the cardiologist-year level. NIH rank is the year-matched ranking of the cardiologist's medical school by total NIH research funding, constructed from the NIH ExPORTER bulk files. Panel A enters rank as $-\log(\text{rank})$. Panel B enters tier indicators with unranked schools as the reference. Column (1) is the rank-only specification. Column (2) adds the training-HRR cath lab share. Column (3) further adds physician characteristics (gender and years since graduation) and practice characteristics (hospital-based share of Medicare claims and log of TIN-level Medicare patient volume). All specifications weight by NSTEMI volume and cluster standard errors at the medical school HRR level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 5. **Effect of Training Environment by Rank Tier**[a]

	Top 25 schools	Below Top 25
Cath lab share, training HRR	0.027	0.038***
	(0.042)	(0.013)
Practice HRR FE	Yes	Yes
Year FE	Yes	Yes
Sample	Top 25	Below Top 25
Observations	1,754	8,831

[a]The outcome is the volume-weighted mean residualized cath rate at the cardiologist-year level. Columns report the within-origin training-imprint specification stratified by the NIH-rank tier of the cardiologist's medical school. The Top 25 sample restricts to cardiologists who attended a medical school ranked in the top 25 by NIH research funding in the matriculation year. Below Top 25 restricts to all other ranked or unranked schools. All specifications include medical school HRR, practice HRR, and year fixed effects, weight by NSTEMI volume, and cluster standard errors at the medical school HRR level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 6. **Effect of Training Environment by Cardiology Subspecialty**[a]

	General Cards	Interventional	EP	Adv HF
Cath lab share, training HRR	0.013	0.058***	-0.142	0.406
	(0.019)	(0.019)	(0.092)	(0.532)
Practice HRR FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	7,570	2,717	358	65

[a]The outcome is the volume-weighted mean residualized cath rate at the cardiologist-year level. Columns report the within-origin training-imprint specification stratified by cardiology subspecialty as recorded in MD-PPAS. General Cards covers cardiologists with the Cardiology primary specialty. Interventional, EP, and Adv HF correspond to Interventional Cardiology, Clinical Cardiac Electrophysiology, and Advanced Heart Failure and Transplant Cardiology, respectively. All specifications include medical school HRR, practice HRR, and year fixed effects, weight by NSTEMI volume, and cluster standard errors at the medical school HRR level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 7. **Effect of Training Environment, Inverse-Probability-Weighted**[a]

	Baseline	IPW
Training-HRR cath lab share	0.058***	0.057***
	(0.021)	(0.021)
Med school HRR FE	Yes	Yes
Practice HRR FE	Yes	Yes
Year FE	Yes	Yes
IPW	No	Yes
Observations	10,729	10,729

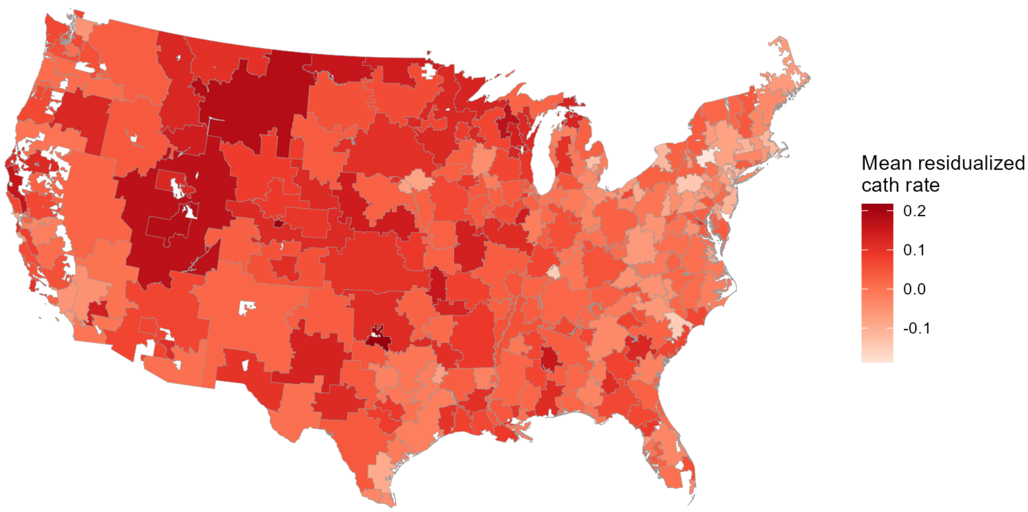
[a]The outcome is the volume-weighted mean residualized cath rate at the cardiologist-year level. Column (1) is the unweighted within-origin baseline (equation 2). Column (2) re-weights cardiologists by the inverse of their estimated propensity to be a mover, where the propensity score is estimated as a function of graduation cohort, gender, subspecialty, and training-period cath lab share. All specifications include medical school HRR, practice HRR, and year fixed effects, and cluster standard errors at the medical school HRR level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 8. **Mid-Career Movers vs. Non-Movers**[a]

	Movers	Non-movers	Diff.	p-value
Training cath share	0.341	0.317	0.024	0.056
Graduation year	1994.4	1993.8	0.7	0.020
Female (%)	7.2	9.1	-1.9	0.101
General Cardiology (%)	67.3	72.9	-5.6	0.006
Interventional (%)	29.3	21.6	7.6	0.000
Electrophysiology (%)	2.4	4.6	-2.2	0.002
Adv. Heart Failure (%)	1.0	0.9	0.2	0.723
Years observed in panel	4.7	3.2	1.5	0.000
Mean NSTEMI volume / year	15.6	14.4	1.3	0.000
Cardiologists	581	5,924		

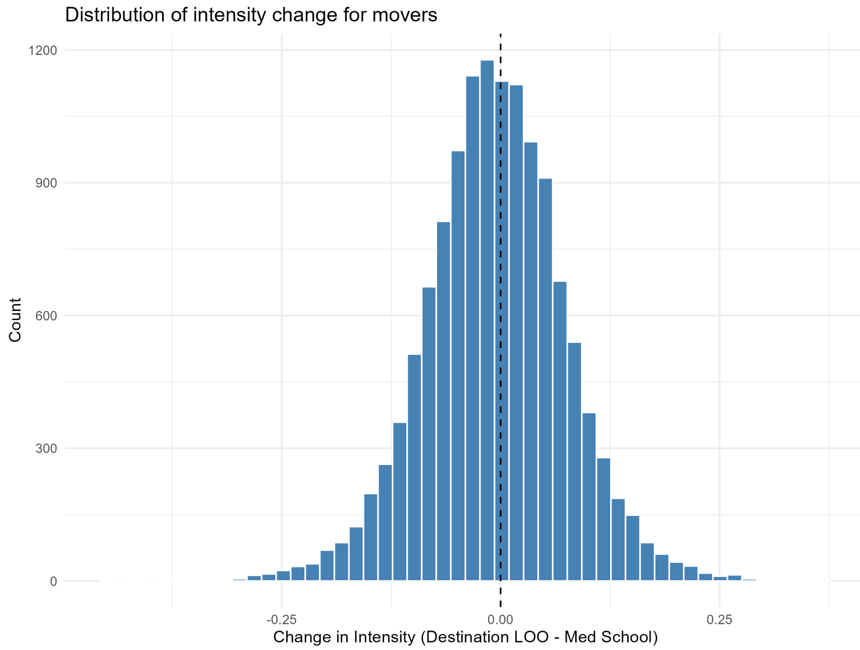
[a]Cardiologist-level comparison of mid-career movers (cardiologists with at least two distinct practice HRRs over the 2008–2018 panel) and non-movers. Means are reported by group, with the difference in means and a p-value from a two-sample t-test. The training cath share is the AHA HRR-year cath lab share at the cardiologist's medical school HRR in the year of matriculation.

Figure 1. **HRR-level Mean Residualized Cath Rate**[a]



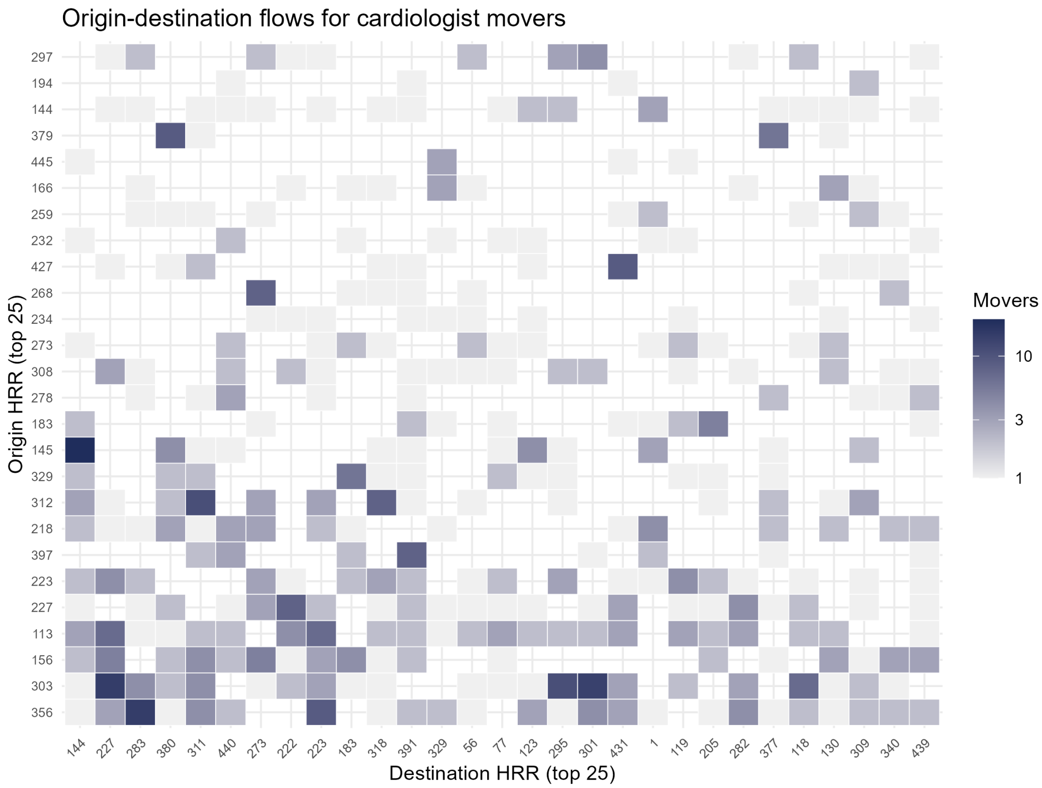
[a]Each polygon is a hospital referral region (HRR). HRRs are shaded by the mean residualized cath rate across cardiologist-years in the analytical sample, with darker shading indicating higher residualized cath rates. The residualized cath rate is the volume-weighted mean of patient-level residuals from a linear probability model regressing an indicator for catheterization within two days of NSTEMI admission on the 31 Elixhauser comorbidities and year fixed effects.

Figure 2. **Distribution of Intensity Change for Movers**[a]



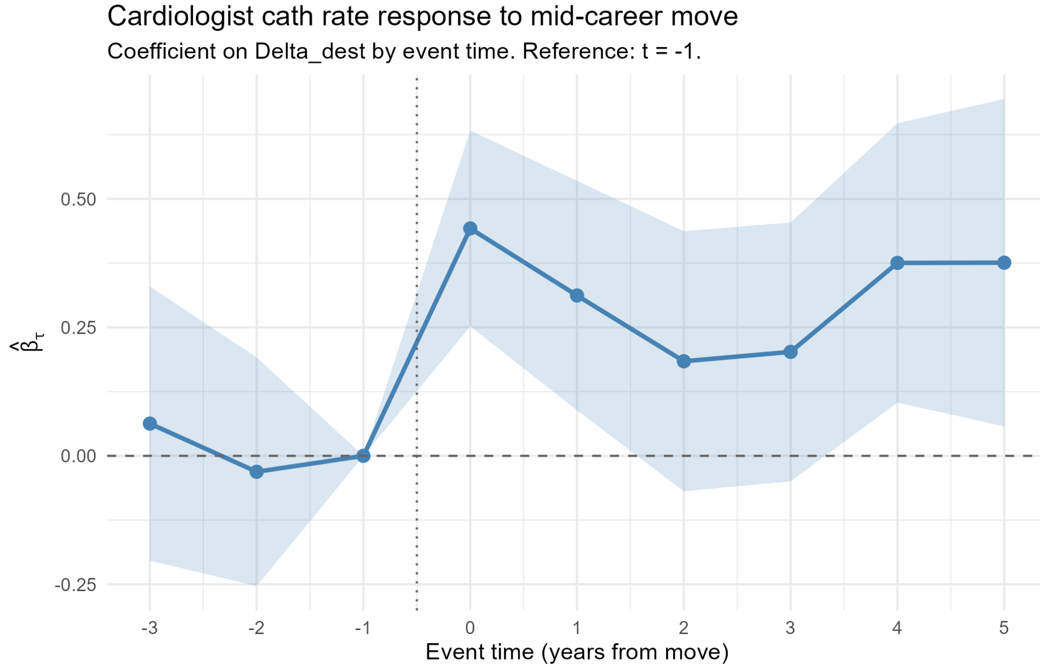
[a]Histogram of the gap, for movers, between the destination peer cath rate (the leave-one-out mean residualized cath rate of other cardiologists in the current practice HRR-year) and the training-period cath lab share at the cardiologist's medical school HRR in the year of matriculation. Movers are cardiologists whose current practice HRR differs from their medical school HRR.

Figure 3. **Origin-Destination Flows, Top 25 Origins and Destinations**[a]



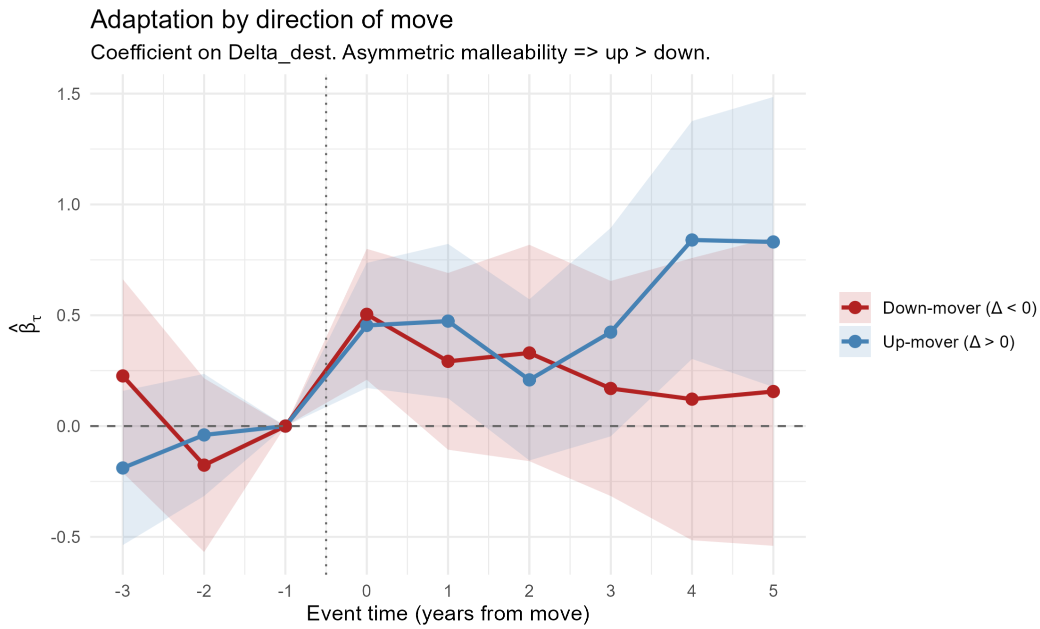
[a]Heatmap of mover counts by medical school HRR (rows) and current practice HRR (columns), restricted to the top 25 origins and top 25 destinations by mover count. Shading intensity indicates the number of movers from each origin HRR to each destination HRR.

Figure 4. **Event Study Around Mid-Career Moves**[a]



[a]Event-study coefficients with 95% confidence intervals from a within-physician specification of the destination response (equation 3). The outcome is the volume-weighted mean residualized cath rate at the cardiologist-year level. Event time τ is measured in years relative to the move year, with the coefficient at $\tau = -1$ normalized to zero. The specification includes cardiologist and calendar-year fixed effects, weights by NSTEMI volume, and clusters standard errors at the cardiologist level.

Figure 5. **Event Study by Direction of Move**[a]



[a]Event-study coefficients with 95% confidence intervals from the within-physician destination-response specification (equation 3), stratified by direction of move. Up-movers are cardiologists whose destination peer cath rate exceeds their origin peer rate at the move year. Down-movers are the reverse. The outcome is the volume-weighted mean residualized cath rate at the cardiologist-year level. The specification includes cardiologist and calendar-year fixed effects, weights by NSTEMI volume, and clusters standard errors at the cardiologist level.